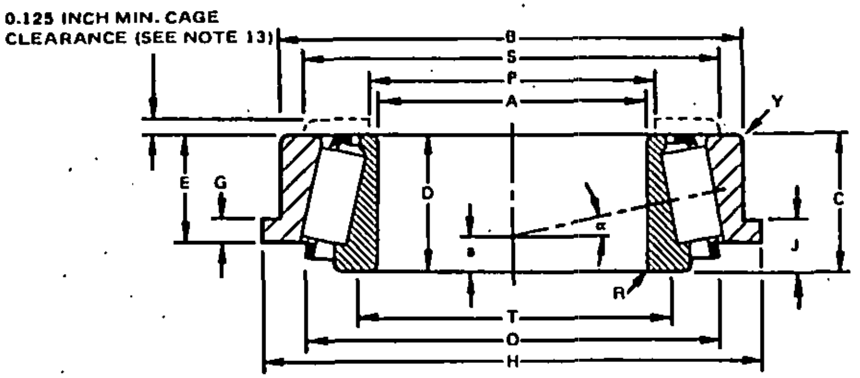
This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document

REVIEWER ACTIVITIES: MI, IS, NS

USER ACTIVITIES: MC, AV, SA

FED. SUP CLASS 3110



Dash No.	AFBMA Ⓐ bearing number (see note 12) Cone – Cup	A	B	C	D	E Ⓐ	J	R	Y	H	G	T	P	S	O	K	Basic dynamic load ratings (lb) Ⓐ		a Ⓐ
		Bore	Outside dia	Bearing width	Cone width	Cup width	Stand-out	Ⓐ Max shaft fillet radius	Ⓐ Max housing fillet radius	Flange		Recommended shoulder diameter				K factor	Radial	Thrust	Effective load center
										Dia	Width	Shaft		Housing					
11	A4050–A4138B	0.5000	1.3775	0.4330	0.4326	0.3437	.1823	.05	.05	1.4985	.0930	0.73	0.67	1.14	1.32	1.29	500	387	–.10
12	A4059–A4138B	0.5906	1.3775	0.4330	0.4326	0.3437	.1823	.03	.05	1.4985	.0930	0.77	0.75	1.14	1.32	1.29	500	387	–.10
1	A6075–A6157B	0.7500	1.5745	0.4730	0.4391	0.3750	.1910	.04	.05	1.6955	.0930	0.94	0.91	1.34	1.50	1.11	530	480	–.06
13	07100–07204B	1.0000	2.0470	0.5910	0.5614	0.5000	.2000	.04	.06	2.1990	.1090	1.20	1.16	1.77	1.97	1.45	1130	780	–.11
2	1780–1729B	1.0000	2.2400	0.7625	0.7810	0.6250	.2935	.03	.03	2.3920	.1560	1.20	1.18	1.95	2.09	1.90	1740	915	–.27
14	17119–17244B	1.1875	2.4409	0.6300	0.6522	0.5625	.2075	.06	.06	2.5930	.1400	1.46	1.36	2.13	2.32	1.53	1620	1060	–.14
3	02476–02420B	1.2500	2.6875	0.8750	0.8750	0.6875	.3437	.03	.06	2.8397	.1562	1.54	1.52	2.32	2.56	1.40	2490	1780	–.20
15	02475–02420B	1.2500	2.6875	0.8750	0.8750	0.6875	.3437	.14	.06	2.8397	.1562	1.75	1.52	2.32	2.56	1.40	2490	1780	–.20
16	2581–2523B	1.3125	2.7500	0.9375	0.9983	0.7500	.3435	.03	.06	2.9020	.1560	1.56	1.54	2.40	2.60	2.14	3280	1530	–.34
17	2585–2523B	1.3125	2.7500	0.9375	0.9983	0.7500	.3435	.14	.06	2.9020	.1560	1.77	1.54	2.40	2.60	2.14	3280	1530	–.34
18	19150–19268B	1.5000	2.6875	0.6250	0.6504	0.4688	.2968	.06	.06	2.8397	.1406	1.77	1.69	2.40	2.64	1.31	2000	1530	–.06
4	3490–3420B	1.5000	3.1250	1.1563	1.1721	0.9375	.4063	.14	.13	3.3090	.1875	2.05	1.81	2.64	2.99	1.60	4150	2590	–.34
19	3382–3320B	1.5625	3.1562	1.1563	1.1965	0.9375	.4063	.14	.13	3.3396	.1875	2.05	1.79	2.76	3.03	2.14	4300	2000	–.43
•5	49175–49368B	1.7500	3.6875	1.2500	1.2500	1.0000	.4688	.14	.13	3.9023	.2188	2.32	2.09	3.23	3.50	1.62	5250	3230	–.36
20	3780–3720B	2.0000	3.6718	1.1875	1.1930	0.9375	.4375	.14	.13	3.8558	.1875	2.52	2.28	3.23	3.54	1.73	5000	2890	–.32

For NOTES see sheets 2 and 3.

Ⓐ denotes changes

APPROVED 30 Apr 1959 REVISED Ⓐ 20 Mar 1978

P. A.	OS	
Other Cust	AT 11	
INTERNATIONAL INTEREST		
TITLE	BEARING, ROLLER, TAPERED, SINGLE ROW OF ROLLERS, NORMAL ANGLE, FLANGED CUP, TYPE 755 (TSF)	
PROCUREMENT SPECIFICATION	FF-B-187	
SUPERSEDES:		

MILITARY STANDARD

MS 19082

SHEET 1 OF 3

3110-0513

DD FORM 672-1

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REVIEWER ACTIVITIES: MI, IS, NS

USER ACTIVITIES: MC, AV, SA

Dash No.	AFBMA Ⓐ bearing number (see note 12)	A	B	C	D	E Ⓐ	J	R	Y	H	G	T	P	S	O	K factor	Basic dynamic load ratings (lb) Ⓐ		a Ⓐ
		Bore	Outside dia	Bearing width	Cone width	Cup width	Stand-out	Ⓐ Max shaft fillet radius	Ⓐ Max housing fillet radius	Flange		Recommended shoulder diameter							Effective load center
	Cone Cup									Dia	Width	Shaft		Housing			Radial	Thrust	
21	529–522B	2.0000	4.0000	1.3750	1.4200	1.0625	.5625	.03	.13	4.2460	.2500	2.32	2.28	3.50	3.82	2.05	6550	3200	–.50
6	529X–522B	2.0000	4.0000	1.3750	1.4200	1.0625	.5625	.14	.13	4.2460	.2500	2.56	2.28	3.50	3.82	2.05	6550	3200	–.50
7	623–612B	2.2500	4.7500	1.6250	1.6250	1.2500	.6562	.14	.13	5.0272	.2812	2.83	2.60	4.13	4.33	1.86	8300	4450	–.55
22	3982–3920B	2.5000	4.4375	1.1875	1.1830	0.9375	.4375	.14	.13	4.6210	.1875	3.03	2.80	3.90	4.25	1.45	5900	4050	–.18
*8	5595–5535B	2.5938	4.8125	1.7188	1.7230	1.4375	.5313	.14	.13	5.0625	.2500	3.27	3.03	4.17	4.65	1.63	10200	6250	–.48
23	39590–39520B	2.6250	4.4375	1.1875	1.1875	0.9375	.4688	.14	.13	4.6523	.2188	3.15	2.91	3.98	4.33	1.72	7000	4050	–.26
24	399A–394AB	2.6875	4.3307	0.8661	0.8660	0.7411	.3120	.09	.05	4.5147	.1870	3.07	2.91	3.98	4.17	1.45	4000	2770	–.03
9	6461–6420B	3.0000	5.8750	2.1250	2.1350	1.7500	.6875	.14	.13	6.1835	.3125	3.78	3.52	5.08	5.51	1.61	14700	9100	–.59
25	594–592B	3.7500	6.0000	1.5625	1.4300	1.1875	.6250	.14	.13	6.2460	.2500	4.33	4.09	5.31	5.67	1.32	9350	7100	–.10
26	687–672B	4.0000	6.6250	1.6250	1.6250	1.1875	.7187	.14	.13	6.9030	.2812	4.65	4.41	5.87	6.30	1.24	11500	9250	–.11
10	861–854B	4.0000	7.5000	2.2500	2.2650	1.7500	.8750	.31	.13	7.8710	.3750	5.08	4.49	6.69	6.85	1.74	19300	11100	–.60
27	56425–56650B	4.2500	6.5000	1.4375	1.4375	1.0625	.6250	.14	.13	6.7460	.2500	4.84	4.61	5.87	6.38	1.18	10100	8600	.08
28	36690–36620B	5.7500	7.6250	1.1250	1.1250	0.9063	.3750	.06	.06	7.7772	.1563	6.10	6.02	7.17	7.48	1.59	9700	6100	.19

*Inactive for new design after 20 Mar 1978.

NOTES:

Ⓐ 1. MATERIAL: Cones, cups, and rollers: bearing quality - carburizing grade alloy or through-hardening grade alloy steel in accordance with FED-STD-66. Cage; carbon steel (one piece stamped).

Ⓐ 2. HARDNESS: Cones, cups and rollers: 58 to 64 Rockwell C.

3. TOLERANCES: Standard grade (AFBMA Class 4) (allowable tolerances are in 0.0001 inch).

Cone bore (A)**			
Size range Ⓐ		Tolerance	
Over	Incl.	Plus	Minus
0	3.0000	5	0
3.0000	6.0000	10	0

Cup diameter (B)			
Size range		Tolerance	
Over	Incl.	Plus	Minus
0.0000	12.0000	10	0
12.0000	24.0000	20	0

Ⓐ Standout (J)			
Bore size range		Tolerance	
Over	Incl.	Plus	Minus
0	4.0000	80	0
4.0000	6.0000	140	100

Ⓐ **Cone bore (A) dimension for dash number 12 is the maximum value; tolerance value is minus by the same magnitude as positive table value.

FED. SUP CLASS 3110

APPROVED 30 Apr 1959 REVISED Ⓐ For changes see pages 1 through 3.

P. A. OS Other Cust AT 11	INTERNATIONAL INTEREST	TITLE BEARING, ROLLER, TAPERED, SINGLE ROW OF ROLLERS, NORMAL ANGLE, FLANGED CUP, TYPE 755 (TSF)	MILITARY STANDARD MS 19082
PROCUREMENT SPECIFICATION FF-B-187	SUPERSEDES:		SHEET 2 OF 3

[illegible] standard is approved for use by all Departments and Agencies of the Department of Defense [illegible] for all [illegible] engineering and design applications and for repetitive use shall be made from this document

REV[illegible] ACTIVITIES MI, IS, NS

USER ACTIVITIES MC, AV, SA

Ⓐ

Cup flange OD (H)			
Size range		Tolerance	
Over	Incl.	Plus	Minus
0	12.0000	20	0

Assembled bearing maximum radial runout		
Cup OD (B)		Tolerance
Over	Incl.	
0	24.0	20

4. DIMENSIONS: All dimensions are in inches. Dimensions T, P, S, and O are recommended shaft and housing shoulder diameters.
Ⓐ Dimensions R and Y are the maximum fillet radii on the shaft and the housing respectively, which will be cleared by the bearing corners.

Ⓐ 5. EFFECTIVE LOAD CENTER: Dimension (a) locates a point on the cone axis which is the center of pressure of all resisting forces set up by the bearing rollers. All moments should be calculated from this point when determining bearing loading and shaft stresses. A minus value of (a) indicates that the center is inside the cone backface.

6. OPERATING TEMPERATURE: Recommended operating temperature not to exceed 121° Celsius (250° Fahrenheit).

Ⓐ 7. LUBRICATION: Bearings shall be furnished without lubrication. Bearings shall be furnished with preservative per MIL-C-11736, Class 3.

Ⓐ 8. BASIC DYNAMIC LOAD RATING: Basic dynamic load rating is that constant stationary load which a group of apparently identical bearings with stationary cups (outer rings) can endure for a rating life of 90 million revolutions of the cone (inner ring). The basic dynamic load ratings listed herein are based on a rated life of 90 million revolutions or 3000 hours at 500 rpm.

Ⓐ 9. RATING LIFE (HOURS): Rating life is the number of hours at some constant speed of the cone (inner ring) that 90 percent of a group of apparently identical bearings will complete or exceed before first evidence of fatigue develops. The magnitude of the rated life in hours is found from the following:

$$L_{10} = \frac{1.5 \times 10^6}{R}\left(\frac{C}{P}\right)^{10/3} \text{ hours}$$

where:

C = Basic dynamic load rating, lb.
P = Equivalent load (combined radial and thrust load), lb.
R = Revolutions per minute.

10. K FACTOR: The K factor is the ratio of the basic radial dynamic load rating to basic thrust dynamic load rating.

11. NORMAL ANGLE: A normal angle bearing has a contact angle (α) between *10 and 19 degrees*. *The contact angle is the angle between* the line of action of the roller load and a plane perpendicular to the bearing axis.

Ⓐ 12. PART NUMBER: The MS part number consists of the MS number, plus the dash number. Example: MS19082-14. The AFBMA (Anti-Friction Bearing Manufacturers Association) cup and cone numbers are for reference only and are not to be used for ordering purposes.

Ⓐ 13. CAGE CLEARANCE: Designers should provide a clearance of .125 inch minimum between the outside edge of the cage and the housing counterbore.

14. For design feature purposes, this standard takes preference over procurement documents referenced herein.

15. Referenced documents shall be of the issue in effect on the date of invitation for bid or request for proposal.

FED. SUP CLASS
3110

APPROVED 30 Apr 1959 REVISED Ⓐ For changes see sheets 1 through 3

P. A. OS
Other Cust
AT
11

INTERNATIONAL INTEREST

TITLE
BEARING, ROLLER, TAPERED, SINGLE ROW OF
ROLLERS, NORMAL ANGLE, FLANGED CUP,
TYPE 755 (TSF)

MILITARY STANDARD
MS 19082

PROCUREMENT SPECIFICATION
FF-B-187

SUPERSEDES:

SHEET 3 OF 3

DD FORM 672-1

3110-0513